A Machine Learning Approach to Understanding Older Adults' Trust in Assistive Robots

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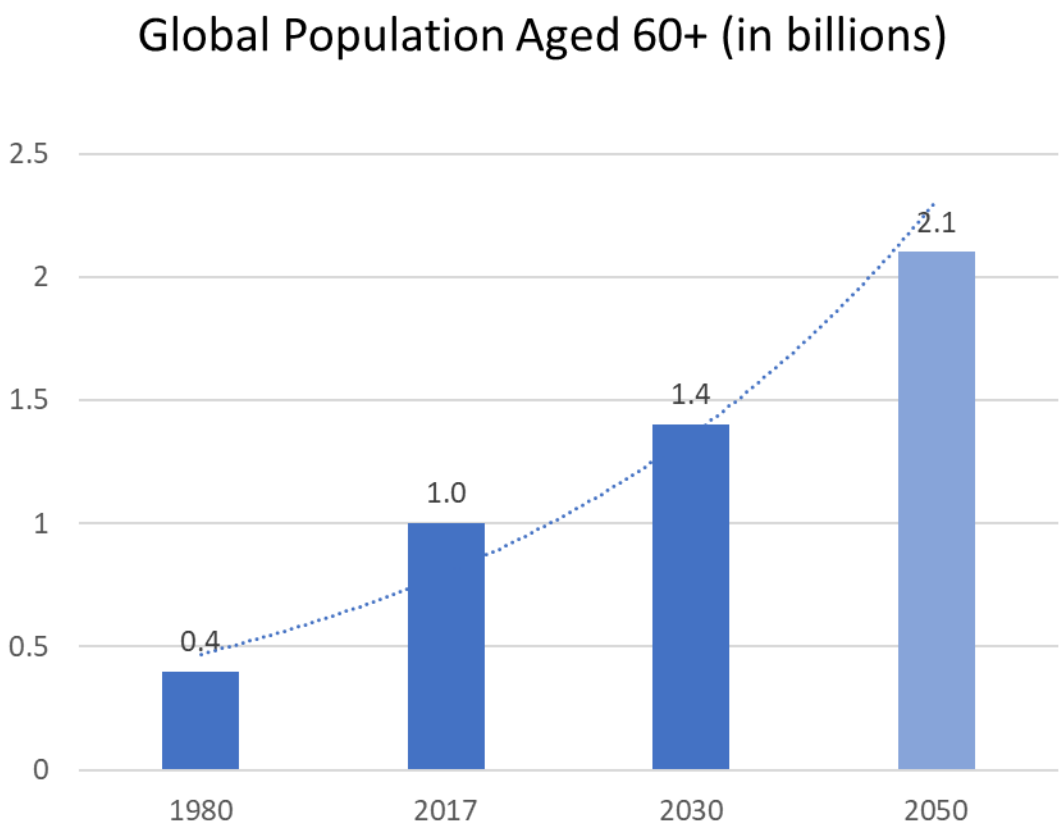
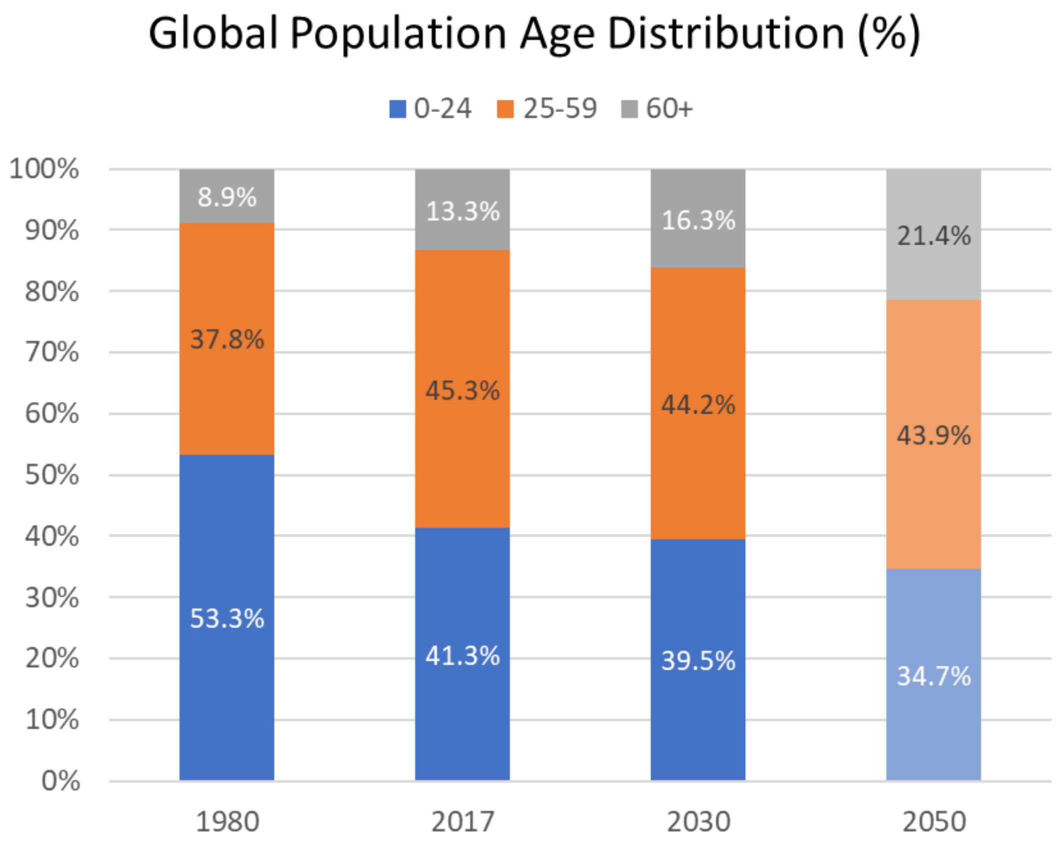
Older Adult Population Trends





United Nations World Population Prospects 2017

Aging in Place

Older Adults prefer to remain in their own homes as they age

What is an Assistive Robot?

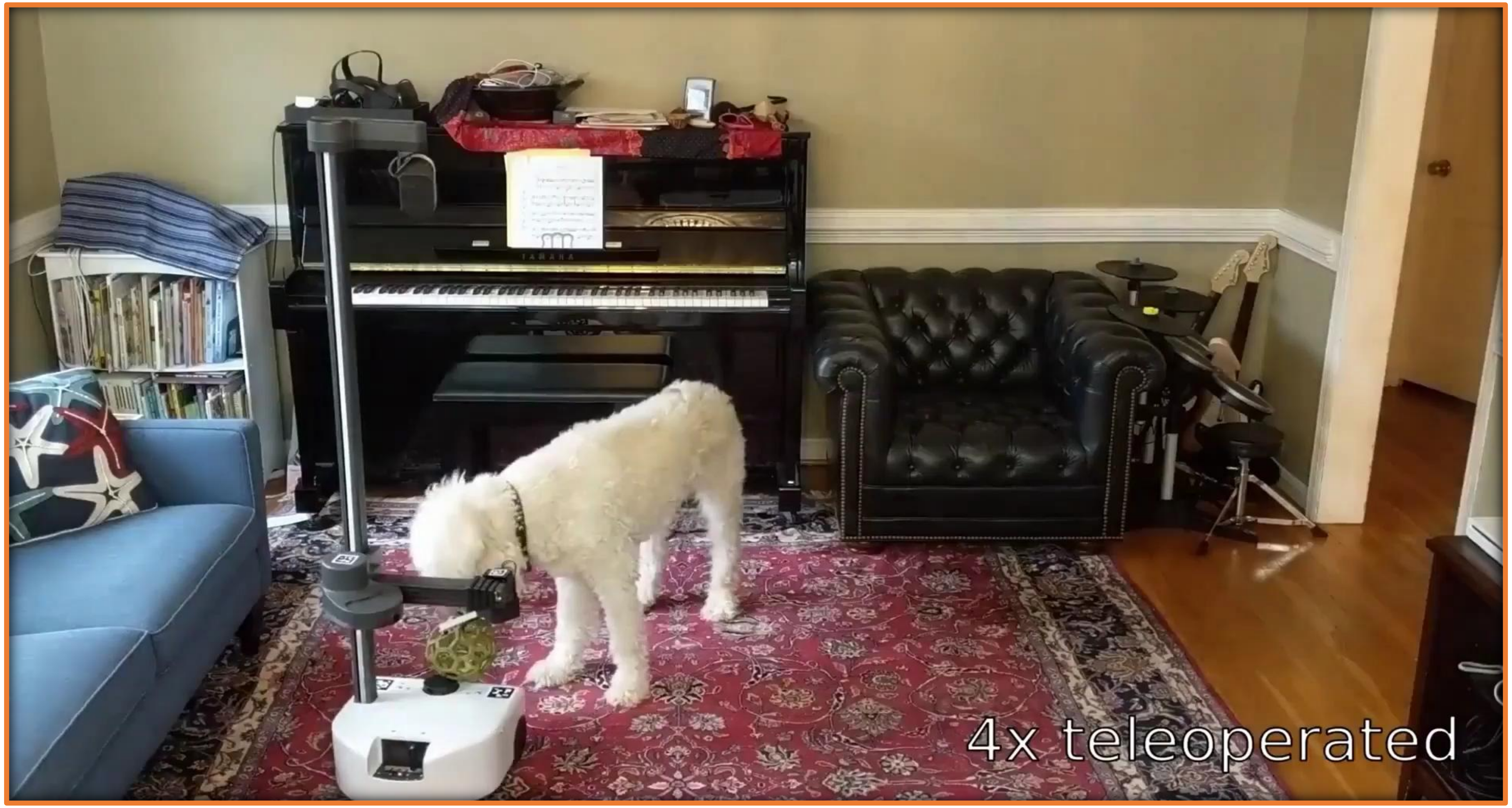
A robot that...
"gives aid or support to a human user"

Feil-Seifer and Mataric, 2005

Applications:

- Cognitive Stimulation
- Safety Monitoring
- Physical Support

Assistive Robots in Action



Challenges of Accepting Assistive Robots

Cost

Usability

Unmet Needs

Trust

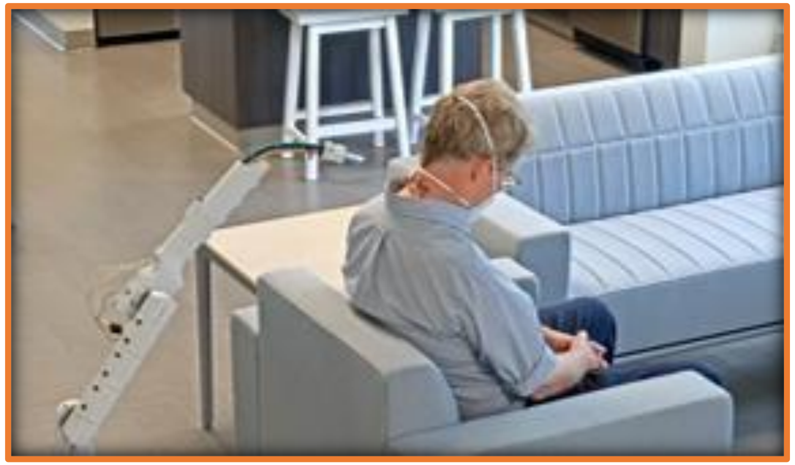
“TRUST:

The degree to which the user accepts to be vulnerable during the interaction, with the confidence that the robot will not harm them”

Primary Research Question

Does older adults' **familiarity** with robotic technologies impact their **trust** in assistive robots?

Assistive Robots in Context



Initial Data (N = 47)

Robot Familiarity and Use

13 different types of robots

(e.g., manufacturing, domestic, entertainment)

Likert Scale of 1-5:

1. Not sure what this robot is
5. Have used or operated this robot frequently

Mitzner et al., 2011

Robot Trust

Evaluating multiple dimensions of trust

(e.g., capability, accuracy, reliability)

Likert Scale of 1-5:

1. Strongly Disagree
5. Strong Agree

Ullman & Malle, 2018

Method

Secondary Data Analysis

Data from user studies exploring the use of assistive robots in different contexts

Machine Learning Approach

Data Analysis Process

- **Identifying suitable models**
  - Least Squares Regression (LSR)
  - LASSO - Least Absolute Shrinkage and Selection Operator (L1)
  - Ridge (L2)

- **Detecting multicollinearity**
  - High VIF indicates variables explain each other

Explanatory Variables (Familiarity with...)	VIF
Autonomous Car	23
Home Robot	20
Toy Robot	32
Manufacturing Robot	94
Military Robot	58
Paro	13
Remote Presence Robot	7
Research Robot	21
Lawn Mower Robot	35
Guard	17
Space Exploration Robot	394
Surgical Robot	191
Unmanned Robot	23

Results

Using LASSO and Ridge Regression Techniques to Predict:

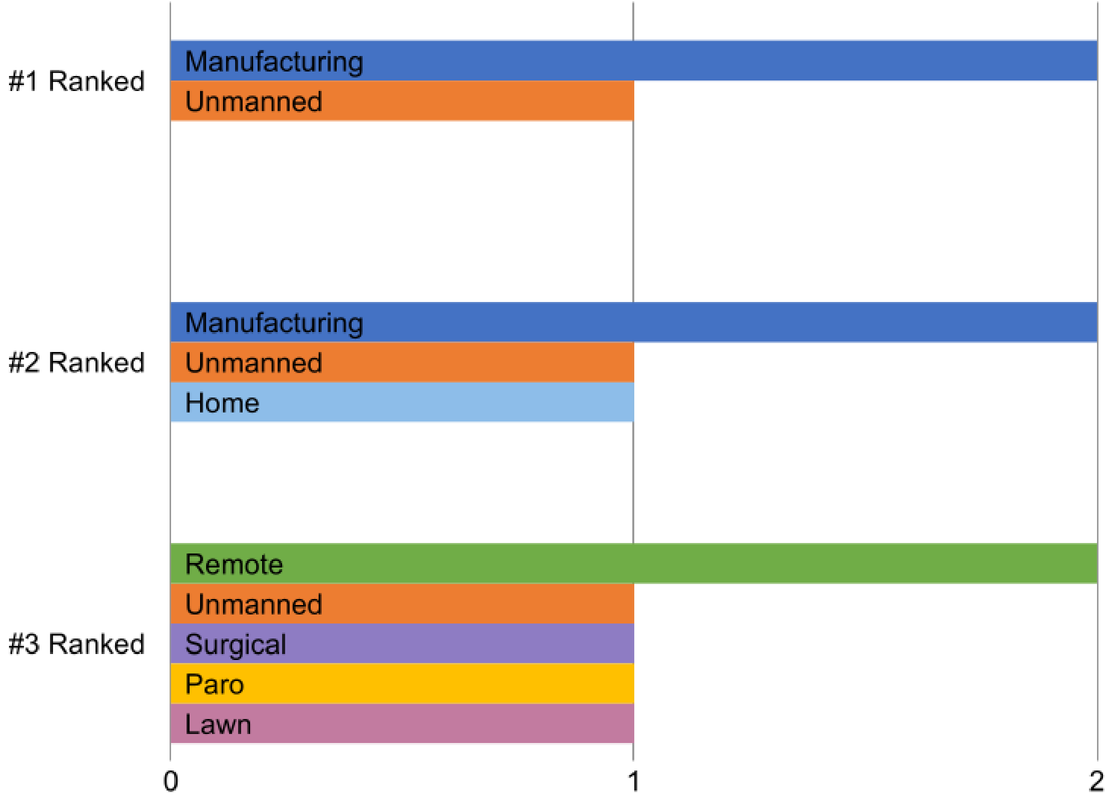
Overall Trust

Capacity-based Trust

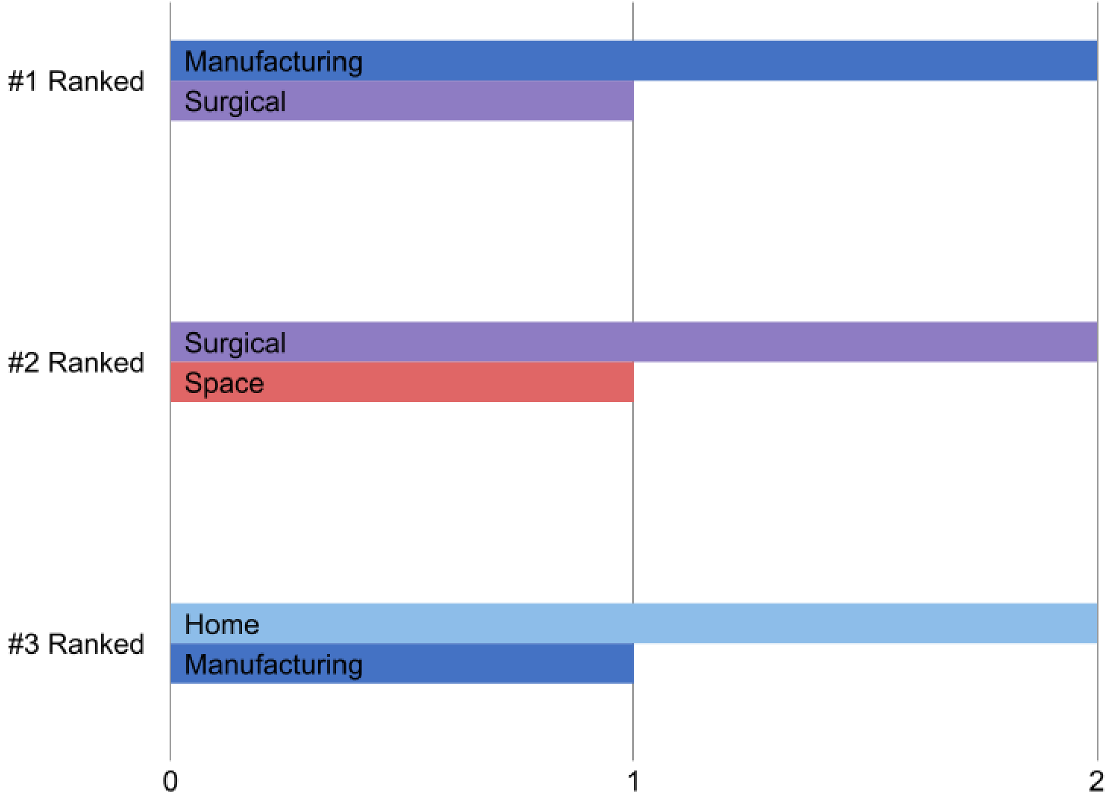
Relation-based Trust

Familiarity with these Robots Impacted Trust

Top 3 Influential Variables from LASSO



Top 3 Influential Variables from Ridge



Discussion

Top predictors of Older Adults' willingness to trust assistive robots include...

Familiarity with:

- Manufacturing robots
- Unmanned robots
- Home robots
- Surgical robots

Surgical Robot

Home Robot: Vacuum

Manufacturing Robot

Next Steps

- Expand dataset
- Data cleaning
- Other ML models
- More demographics
- Factors beyond robot familiarity
- Better measures of trust (qualitative + quantitative)

Concluding Thoughts

Engagement in participatory design...

builds **familiarity** ➡ identifies gaps in **trust**

Increasing awareness of assistive tech could...

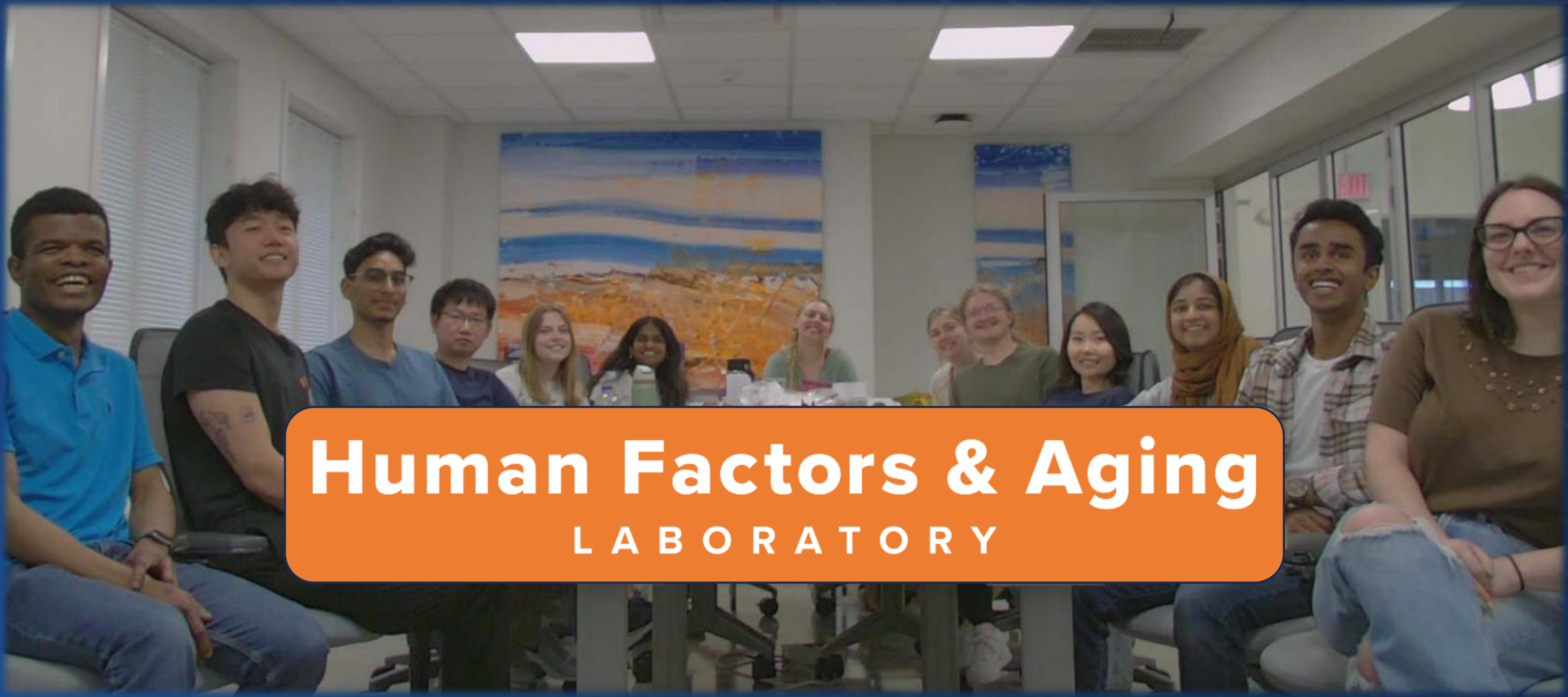
improve **trust** ➡ promote user **acceptance**

Designing robot communication with users' familiarity in mind can help foster trust and accelerate acceptance during interactions.

Overarching Goal

A future in which older adults **trust** assistive robots

Thank You!



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Supplementary Videos

The following slides show 3 assistive robots across various use-cases.


4x teleoperated


LOVE